CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 2: **Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area (in square units) of the triangle formed by the line $y = 2x + 10$ and the two axes is: 1

(a) 25 **(b)** 27 **(c)** 26 **(d)** 24

2. The x-coordinate of the point where $y = 5x + 1$ and $y = 2x + 13$ meet is: 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 6

3. If $p(x) = 2x + 3$, then $p(p(4)) =$ 1

(a) 25 **(b)** 24 **(c)** 26 **(d)** 27

4. A plan costs ₹100 plus ₹20 per unit; another costs ₹220 plus ₹10 per unit. They cost the same for ___ units. 1

(a) 12 **(b)** 14 **(c)** 13 **(d)** 11

5. If $x = 4$ is the zero of $kx - 28$, then $k =$ 1

(a) 7 **(b)** 8 **(c)** 9 **(d)** 6

6. If a linear polynomial $ax + b$ has zero 0, then: 1

(a) $a = 0$ **(b)** $a = b$ **(c)** $b = 0$ **(d)** $b = 1$

7. For the pattern with nth term $6n - 1$, the sum of the 1st and 10th terms is: 1

(a) 66 **(b)** 63 **(c)** 65 **(d)** 64

8. For which value of n is $13n - 4 = 217$? 1

(a) 19 **(b)** 18 **(c)** 16 **(d)** 17

9. **Assertion (A):** The zero of the polynomial $p(x) = 3x - 12$ is 4. 1

Reason (R): The zero of $ax + b$ ($a \neq 0$) is $-\dfrac{b}{a}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The graph of $y = 2x - 6$ cuts the x-axis at $(3, 0)$. 1
Reason (R): The graph of $y = 2x - 6$ cuts the y-axis at $(0, -6)$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Plan A costs ₹200 plus ₹30 per unit; Plan B costs ₹440 plus ₹10 per unit. For how many units do both cost the same? 2

12. If $p(x) = 3x + 4$, find $p(p(x))$ and its value at $x = 2$. 2

13. Find the area of the triangle formed with the axes by the line $y = 2x + 14$. 2

14. Plan A costs ₹150 plus ₹25 per unit; Plan B costs ₹390 plus ₹5 per unit. For how many units do both cost the same? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the point of intersection of $y = 5x + 2$ and $y = 2x + 11$. 3

16. A linear polynomial has $p(1) = 12$ and $p(3) = 20$. Find $p(x)$, its zero and $p(6)$. 3

17. The nth term of a pattern is $14n + 3$. Find the number of terms that are less than 143 and the sum of the first two terms. 3

OR

Find the point of intersection of $y = 7x - 3$ and $y = 3x + 9$. 3

18. A linear polynomial has $p(1) = -10$ and $p(3) = 0$. Find $p(x)$, its zero and $p(4)$. 3

OR

The nth term of a pattern is $11n - 4$. Find the number of terms that are less than 128 and the sum of the first two terms. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A taxi company in Jaipur charges a fixed amount of Rs 50 plus Rs 12 for every kilometre travelled. The fare for x km is given by a polynomial $C(x)$. 5
(i) Write $C(x)$.
(ii) Find the fare for a 10 km ride.
(iii) For what distance is the fare Rs 230?
(iv) State the slope and the y-intercept of the graph of $y = C(x)$ and say what each means.

20. Two taxi services: A charges ₹30 plus ₹12 per km; B charges ₹90 plus ₹7 per km. Find the distance for which both charge equally, the common fare, and which is cheaper for 8 km. 5

OR

A line passes through $(2, 0)$ and $(6, 20)$. Find its slope, equation, intercepts and the area of the triangle it forms with the axes. 5

END OF QUESTION PAPER — **Total: 40 Marks**